

Khanh Nguyen

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SUMMARY

Machine Learning engineer with 7+ years building and operating production-grade ML systems at scale — NLP, speech, and multimodal — from data pipelines and model training to low-latency, cost-bounded inference. Built LLM-agent and MCP tooling that puts an agentic layer over complex engineering workflows, owns CI/CD and Kubernetes deployment for models serving hundreds of thousands of daily requests, and builds the developer-facing interfaces that drive adoption.

TECHNICAL SKILLS

ML & Inference: PyTorch, TensorFlow, ONNX, vLLM, LLM agents & tool use (MCP), RAG, NLP, speech (ASR), computer vision / OCR, model quantization & inference optimization (CTranslate2, 8-bit, dynamic batching), Weights & Biases

Delivery & Infra: CI/CD (GitLab), Docker, Kubernetes (distributed training & deployment), AWS (S3), GCP, production ML deployment, observability & monitoring

Data & Pipelines: Large-scale data processing, dataset curation, crowd annotation & labeling workflows, preprocessing & filtering, vector indexing (FAISS), multimodal data (text / speech / image / OCR)

Languages & Backend: Python (expert), TypeScript, FastAPI, Flask, Node.js, REST API design, system design, SQL

PROFESSIONAL EXPERIENCE

Senior Software Engineer — QuantrolOx — Espoo, Finland 07/2022 – Present

- Built an LLM agent (served with vLLM) and MCP server as the agentic layer over complex hardware and experiment workflows, enabling natural-language control and automation of multi-step engineering processes.
- Built production ML systems for automated device calibration, reaching 99.8% gate fidelity through autonomous optimization workflows that cut calibration from ~1 day of manual tuning to ~30 minutes.
- Designed and built a customer-facing Python SDK and internal tooling that became the primary developer interface for composing custom workflows, driving adoption across 2 research teams.
- Built real-time data processing and analysis pipelines for experiments — raw ingestion, transformation, and analysis — with React/TypeScript dashboards for monitoring.

Machine Learning Engineer — Vainu — Helsinki, Finland 05/2018 – 02/2022

- Trained and deployed production NLP systems as AI services serving hundreds of thousands of daily requests at ~95% accuracy across 7 languages (PyTorch, TensorFlow).
- Optimized model inference for low-latency, cost-bounded serving with dynamic / adaptive batching, cutting memory usage 40% while sustaining high request throughput.
- Owned ML infrastructure on AWS (S3) and Kubernetes with automated model deployment via GitLab CI/CD, orchestrating distributed training workloads and cutting deployment time from 4 hours to 10 minutes.
- Ran crowd annotation workflows end to end: designed labeling tasks and curated ~30K labeled examples into model-ready training sets.

Software Engineer / Data Scientist — Swappie — Helsinki, Finland 02/2022 – 07/2022

- Built an end-to-end ML pipeline on Google Cloud (TensorFlow, FastAPI) and deployed demand-forecasting models from rapid prototyping to production, translating business requirements into ML solutions with product teams.

Fullstack Developer — RND Works — Vantaa, Finland 01/2017 – 05/2018

- Trained the first Finnish word embeddings and built a native Finnish chatbot with Google Cloud speech/vision integration (TensorFlow, Flask).

SELECTED PROJECTS

Speech-to-Text Inference Pipeline

- Profiled and optimized model inference with ONNX, CTranslate2, and 8-bit quantization for low-latency, cost-bounded transcription; deployed as a FastAPI service.
- Built a Conformer-based ASR pipeline handling regional dialect variation, including data preprocessing and training-sample preparation from raw audio.

Invoice Processing with Computer Vision

- Built a multimodal document-understanding pipeline combining OCR (Tesseract) and local LLMs for automated extraction, with image preprocessing across diverse PDF/image formats and privacy-preserving local deployment.

Vietnamese Legal RAG Assistant

- Built an end-to-end retrieval pipeline (PyTorch, FastAPI, FAISS, BERT embeddings): document ingestion, vector indexing, and a user-facing chat interface.

EDUCATION & LANGUAGES

Master of Applied Data Science — University of Michigan, USA · 2024 · GPA 3.8 · Focus: ML, computer vision, deep learning, production systems
English (Fluent) · Vietnamese (Native) · Finnish (Conversational)